ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE F · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice f.

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How to mark: each ✓ is one mark. Accept other correct answers, sensible project decisions and neat, recognisable sketches, tables and circuits.

QUESTION 1

1.1 A design brief is a short statement of the problem and what the client needs ✓✓ (e.g. "an affordable machine to fill juice bottles automatically"). A design specification is a detailed list of the exact requirements and limits that the solution must meet ✓✓ (output per hour, size, cost, materials, safety). (4)

1.2 Any THREE ✓✓✓: budget; available floor space; time available; food-safety regulations; available power and compressed air; bottle sizes; skills of the staff. (3)

1.3 The machine must be safe and easy for them to operate ✓, clean and maintain with their skills ✓; otherwise it will be misused, break down or not be accepted ✓. (3)

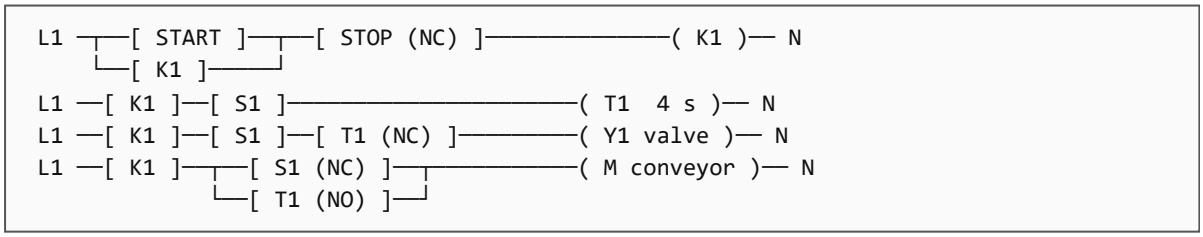
[10]

QUESTION 2

2.1 Neat sketch ✓ showing: conveyor and drive motor ✓; bottle sensor at the nozzle ✓; nozzle with valve under the tank ✓; labels ✓ (5)

2.2 Steps in a logical order ✓✓✓✓✓, e.g.: (1) press START – conveyor runs; (2) bottle moves towards the nozzle; (3) sensor S1 detects the bottle; (4) conveyor stops; (5) valve Y1 opens and the timer runs for 4 s; (6) valve closes and the conveyor restarts, moving the full bottle away; (7) repeat for the next bottle until STOP is pressed. (6)

2.3 START / STOP latching circuit with relay K1 ✓✓; S1 (NO) with a K1 contact drives timer T1 ✓; Y1 on through S1 and NC T1 ✓✓; motor M through K1 and an NC S1 contact (stops while a bottle is under the nozzle) ✓; T1 done contact restarts the conveyor (NO T1 in parallel with NC S1) ✓; neat and correct symbols ✓, e.g. (8)



2.4 Any FOUR ✓✓✓✓: photoelectric (through-beam, retro-reflective or diffuse); capacitive proximity sensor; ultrasonic sensor; mechanical limit switch with a light lever; fibre-optic sensor; vision camera. (4)

2.5 It does not rust or react with the acidic juice ✓, and it is smooth and easy to clean, so it meets food-hygiene requirements ✓. (2)

[25]

QUESTION 3

3.1 Project manager – plans the work, budget and schedule, and leads the team ✓✓; designer – designs the machine and produces the drawings and specifications ✓✓; technician – builds, wires, programs and tests the machine ✓✓; quality controller – checks that the parts and the finished machine meet the specifications and standards ✓✓. (8)

3.2 Suitable headings ✓✓✓ (item, description, quantity, unit cost, total cost, supplier); at least FIVE sensible items ✓✓✓ (e.g. conveyor, motor, PLC, sensor, solenoid valve, stainless nozzle, frame, wiring); a total / contingency row ✓; neat table ✓ (8)

3.3 A part (detail) drawing shows ONE component with all its dimensions, tolerances and material so it can be made ✓✓. An exploded drawing shows how all the parts fit together, with the parts drawn slightly apart in assembly order and numbered ✓✓. (4)

3.4 Any FIVE ✓✓✓✓✓: wear PPE (safety shoes, glasses, gloves); keep the power and air isolated and locked out; use the correct tools in good condition; lift heavy parts correctly or with help; keep the work area clean and tidy; secure the frame before working on it; follow the drawings and instructions; be careful with sharp edges. (5)

3.5 Procurement is the process of finding suppliers and buying the materials and parts for the project ✓✓. Lead time is the time between ordering an item and receiving it ✓✓; parts with long lead times must be ordered early, or the whole project will be delayed ✓. (5)

[30]

QUESTION 4

4.1 Functional testing checks that every part of the machine does what it should (sensor detects, valve opens, conveyor stops) ✓✓. Performance testing checks how well it works against the specification – speed, accuracy of the fill volume, bottles per hour, reliability over time ✓✓. (4)

4.2 Any THREE ✓✓✓: juice leaks or spills; bottles falling over or jamming; the valve not closing; a sensor not detecting; a safety device failing; motor overheating; abnormal noise. (3)

4.3 Errors in the logic can be found and corrected safely ✓ before they can damage the machine, waste product or injure someone ✓. (2)

4.4 Fill a few bottles and measure the volume in each ✓; change the timer preset in small steps (longer if under-filled, shorter if over-filled) ✓; repeat until the volume is correct and consistent ✓. (3)

4.5 It is proof that the machine was tested and works as agreed in the specification ✓; it shows that the client accepts the machine ✓, and from then on is responsible for its operation (and the warranty starts) ✓. (3)

[15]

QUESTION 5

5.1 Four items, each with a task ✓ and a frequency ✓, e.g. nozzle and valve – clean and sanitise – daily; conveyor belt – check tension and tracking – weekly; motor and gearbox – check for noise, heat and oil – monthly; sensor – clean the lens and test – daily; air / electrical connections – check for leaks and tightness – monthly. (8)

5.2 Place a bottle (or object) in front of the sensor ✓ and check that the sensor's LED comes on ✓ and that the matching PLC input LED / status in the software changes ✓. (3)

5.3 Any FOUR ✓✓✓✓: introduction and description; safety information; installation and commissioning; operating instructions; maintenance schedule; troubleshooting guide; circuit and pneumatic diagrams; parts list. (4)

5.4 A break in a circuit so that current cannot flow (e.g. a broken wire or loose terminal) ✓✓; found with a continuity test / resistance measurement or by checking the voltage at points along the circuit ✓. (3)

5.5 Any ONE ✓✓: faults are reported immediately even when no one is nearby; production data can be checked from anywhere; maintenance can be planned before a breakdown. (2)

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TOTAL: 100

END OF MARKING GUIDELINE